6MMCS009W1 Usability Testing & Evaluation

Individual Coursework

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1 Introduction

With the prevailing trend in modern economy a business must identify the needs of customers and deliver an extraordinary online shopping experience, thus UX design plays a crucial role in the contemporary e-business. With the growth of e-tail, more and more companies are beginning to realize that having a good UI design not only means better user satisfaction, easier to navigate sites, but increased sales . In this report, the subject is an online retail business venture considering revamping its e-shop UI to prompt better ergonomics and interactivity.

This paper aims at assessing design options that were produced by a UX design firm, which generated four design versions with three different shades per version and three type sizes. All of these designs were tested with the actual users across the targeted device platforms, namely, the desktop, mobile, and tablet versions. Learnability, satisfaction levels, number of clicks, engagement scores, and errors will be collected to support recommendations for the optimal design according to scientific research.

I believe that perhaps the biggest advantage of UX research in e-commerce is that it employs both qualitative and quantitative data gathering processes to determine customers' needs and issues associated with their experiences. A good usability design should not only beautify a website but also reduce features that may interfere with the purchase process of a particular product. This calls for design of a navigation system that will be readily accessible by all the categories of user including the persons with disability or those that are technically challenged. Furthermore, a good attitude towards an application makes users to repeat their purchases, meaning more revenue for the firm.

In light of these considerations, this report will analyze two specific research questions related to usability testing metrics: the time it takes to familiarise with the interface between two design variations for mobile users and the variation in perceived satisfaction among the different device type. This research will use the Likert scale and paired sample t-tests and ANOVA will be used to examine which design aspects are most appealing to users across the multiple platforms. In conclusion, these findings will enable the client to make some strategic choices about redesigning the UI of the e-shop for better assurance of its efficiency and relevancy now and in the near future, based on the customers' changing expectations.

2 Hypothesis testing

The following section identifies two of the RQs from the study and details the corresponding independent and dependent variables, usability metrics, and statistical testing methods used in the analysis of data. Each of the research questions was designed to find answers to a different aspect of user experience regarding the e-shop interface of the online retail business.

In this study Research Question 1 is Learnability of Designs A and C for Mobile Users,

RQ1: Does the time required to learn the interface differ between Design A and Design C for mobile users?

- **Independent Variable:** The design variations being compared are Design A and Design C.
- **Dependent Variable:** The primary metric of interest is the time taken by users to learn how to complete a task using each design.
- **Usability Metric Studied:** Learnability, which assesses how quickly users can become proficient with a new interface.
- **Testing Method:** A paired sample t-test will be employed to compare the means of the time taken to learn between the two designs. This statistical method is appropriate here as it allows for the comparison of two related groups—users who interacted with both Design A and Design C. By analyzing the differences in learning times, we can determine if there is a statistically significant difference in learnability between these two designs.

In this study Research Question 2 is Satisfaction Levels Across Device Types

RQ2: Does the satisfaction level differ across different devices?

- **Independent Variable:** The type of device used by participants, categorized as Desktop, Mobile, or Tablet.
- **Dependent Variable:** The number of clicks made by users when interacting with Design A, which serves as a proxy for user satisfaction.
- **Usability Metric Studied:** Satisfaction levels assessed through user interactions, specifically focusing on how many clicks were needed to complete tasks.

- **Testing Method:** ANOVA (Analysis of Variance) will be utilized to compare satisfaction levels across multiple groups (the three device types). ANOVA is suitable for this scenario because it allows us to determine if there are any statistically significant differences in means among three or more independent groups. This analysis will help identify whether device type influences user satisfaction when interacting with Design A.

2.1 Paired sample t-test

2.1.1 Hypothesis testing for Paired sample t-test

Research question	*Does the time required to learn the interface differ between Design A and Design C for Mobile users?*
Hypothesis statement	***Null Hypothesis (H_0):*** There is no significant difference in the time required to learn Design A compared to Design C for mobile users.
	Alternative Hypothesis (H_a): There is a significant difference in the time required to learn Design A compared to Design C for mobile users.
Mathematical expression of the Hypothesis	***Null Hypothesis (H_0):*** There is no significant difference in the time required to learn Design A compared to Design C for mobile users. $H_0: \mu_A = \mu_C$
	Alternative Hypothesis (H_a): There is a significant difference in the time required to learn Design A compared to Design C for mobile users. $Ha: \mu_A \neq \mu_C$
Dependent variable	The time to learn
Dependent variable data type	Ratio
Independent variable	Design (A vs C)
Independent variable type	Nominal

Control variable	• User experience level • Task Complexity • Device Type
Experimental design	Within subject design
Study method	Paired Sample T test

2.1.2 Data organisation

When organizing data for analysis in a specific statistical tool f SPSS the procedure involves arranging for data file in statistical tool. The first step of this process involves importing the contained dataset and the determination of those columns key in answering the posed research questions. For the current study, we focus on three specific columns: The three groups were: “Device Type”, “Learn to complete a task with Design A”, and “Learn to complete a task with Design C.” The other columns are considered as irrelevant to this investigation and thus are excluded to make the dataset more convenient.

Data Organization Steps as this,

- **Importing Data:** The data set is brought into the SPSS so that the entries will be formatted and understandable by the program.
- **Column Selection:** Subsequent to data importation we tend to delete all other columns that may not be relevant to the analysis. This helps in preventing mental blurring of the various factors into one big blob of data which can be really hard to sort out.
- **Variable Definition in Variable View:** In SPSS, we switch to "Variable View" to define each variable appropriately: Device Type is a nominal variable which segments users by device type which could be either Desktop, Mobile or Tablet. Learn to Complete a Task with - Design A is a ratio (interval) variable where the independent variable is the time taken (measured in seconds) by the users to effectively learn on how to perform certain tasks on Design A. Learn to Complete a Task with - Design C is also a type of extraneous variable like in Design A but a continuous (ratio) variable that measures the time taken for Design C.

- Data Cleaning: However, they also said that before moving to the analysis, it is obligatory to refine the received data. Missing Values: Some of the data points that are missing are as well noted. Regarding the analysis requirements, these can be answered by either excluding the cases or by using the imputations by means of the mean or median equations. Outliers: Outliers are dealt with by use of a box plot or descriptive statistics. This means that if there is any outlier in the data, it may be omitted or followed up to try and find out why it is different. Consistency Checks: During the stage, the dataset is screened for any anomalies in entry or format (dates, times, etc., that enter the wrong format, or values that are grossly unreasonable such as a negative time).
- Creating Computed Variables: To assist in analysis, computed variables can be created as needed. For instance, if we wanted to find an average learning time across designs for each participant, we can create a new variable.
- Data View Data Arrangement: Having defined and cleaned all the variables, we now arrange the data in "Data View" according to the needs of our statistical tests. Make sure that each row represents one participant's responses. Columns should only include those variables relevant for the paired sample t-test analysis.

Analysis Preparation After organizing and cleaning the data, we proceed to conduct descriptive statistics and visualizations: To obtain descriptive statistics and graphs, navigate to the "Analyze" tab in SPSS: Choose ‘Statistics’ then ‘Explore’. As independent variable take “Device Type” and make “Learn to Complete a Task with - Design A” and “Learn to Complete a Task with - Design C” as dependent variables. After that if you want to get printable output tables and Graphs with means, SD (standard deviations) & distribution on click “OK”.

Statistical Testing Following descriptive analysis, we will perform hypothesis testing: You will again be under the “Analyze” tab, choose “Compare Means” then “Paired-Samples T Test.” Select the pair: Learn to Complete a Task with – Design A and Learn to Complete a Task with – Design C. Perform the test to compare the means of learning times of mobile users with the two designs of the sites to determine if one design is statistically faster than the other. Following such structured process, we ensure our data is clean and ready for statistical testing and therefore give accurate conclusions on overall user experience across the various design interface.

2.1.3 Descriptive statistics and discussion of results

In this section is to give a full analysis of the descriptive statistics concerning the time taken to learn a task using two designs- Design A and Design C. Descriptive statistics describe the central tendency, variability, and distribution of the data, enabling one to understand how participants performed under each design. This is followed by a discussion on the measures of central tendency such as mean, median, variance, standard deviation, range, and interquartile range for the two designs in an attempt to contrast the differences between the two on the task completion time. It also presents skewness and kurtosis values for the normality of the data to ascertain whether to apply parametric or non-parametric statistical methods in investigating whether findings are significant.

Descriptives

			Statistic	Std. Error
learn to complete a task with - Design A	Mean		25.28	.289
	95% Confidence Interval for Mean	Lower Bound	24.69	
		Upper Bound	25.86	
	5% Trimmed Mean		25.31	
	Median		25.50	
	Variance		3.333	
	Std. Deviation		1.826	
	Minimum		22	
	Maximum		28	
	Range		6	
	Interquartile Range		3	
	Skewness		-.431	.374
	Kurtosis		-.820	.733
learn to complete a task with - Design C	Mean		27.28	.476
	95% Confidence Interval for Mean	Lower Bound	26.31	
		Upper Bound	28.24	
	5% Trimmed Mean		27.31	
	Median		28.00	
	Variance		9.076	
	Std. Deviation		3.013	
	Minimum		21	
	Maximum		33	
	Range		12	
	Interquartile Range		5	
	Skewness		-.160	.374
	Kurtosis		-.617	.733

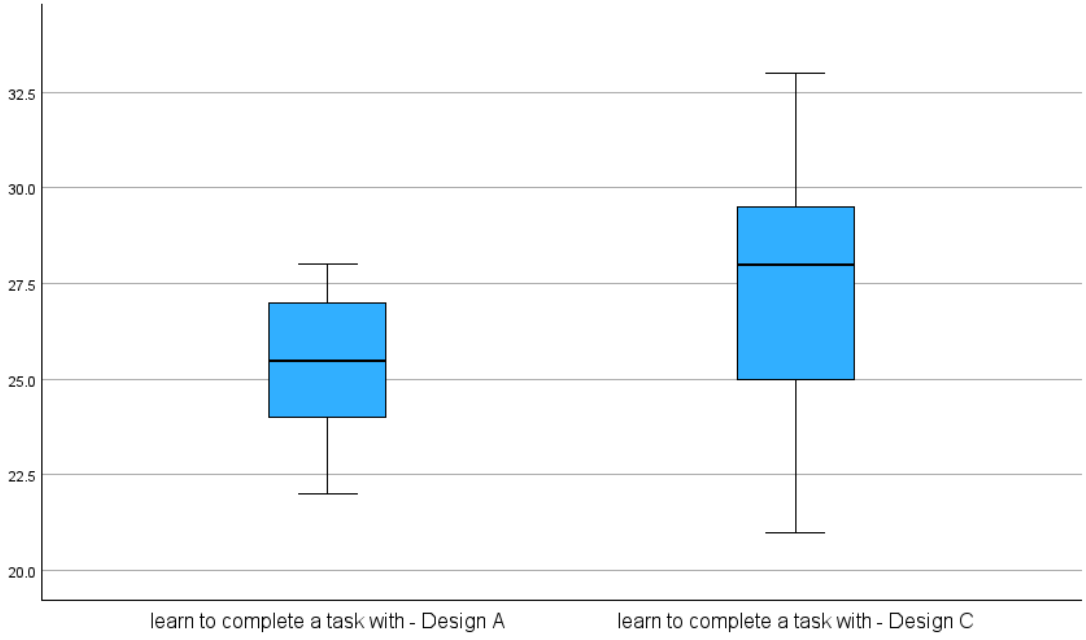
1. Central Tendency (Mean, Median, Trimmed Mean):
    - The mean learning time of a task is also higher for Design C (27.28) compared to Design A (25.28) implying that participants were slower with Design C than with Design A.
    - For Design C it is 28 and for Design A it is 25.50, which means the actual time which has been spent also is more for Design C.
    - The 5% trimmed mean is nearly equal to the actual mean for both Design C and Design A designs and thus, reveals that the real mean in both designs has not been affected by the outliers to any extent.

2. Dispersion (Variance, Standard Deviation, Range, Interquartile Range):
    - The variance and the standard deviation of task completion time: Hence, the variance of Design C (9.076) was higher from Design A (3.333); hence, the variability was more in the Design C with SD (3.013) relatively higher than Design A (1.826).
    - Both designs have different interquartile ranges (IQR): To assess the dispersion, the designs have the follow median absolute deviations: MDAD = 5 for Design C and MDAD = 3 for Design A, which proves that the mid 5090 50 percentile range is distributed more broadly for Design C.
    - Again Design C has more variability in the total time range than Design A; it varies from 21 to 33 while Design A varies from 22 to 28.

3. Skewness and Kurtosis:
    - Skewness: As we have seen, Design A has a skew value of -0.431 this being an evidence of a left skew, that is, the distribution of the variables involved is not perfectly normal. Distribution of Design C indicates moderate left skewed hence skewness of -1.160.
    - Interpretation: It will be noted both distributions are fairly ‘asymmetrical’ with ‘skewed’ orientation, though the skewness is slightly higher in Design C.

- Kurtosis: In design A the kurtosis = -0.820 and in design C the kurtosis = -0.617, which in general both distributions are Platykurtic because they have fatter tails to the Normal Kurtosis value.

Interpretation: Following this, as we can see both designs have less number of outliners and Design A is slightly more flatter than Design C.



In this boxplot provided visually compares the distribution of task completion times for Design A and Design C.

For Design A

Median (Central Line in the Box): The average task completion time for Design A is 25.5, meaning that the respective test must have been passed by the half of the population or more of participants In less than 25.5 seconds.

Interquartile Range (IQR): The box depicts the so-called Interquartile range or IQR, which extend from the first quartile, Q1 to the third quartile, Q3. This range represents the middle 50% of Design A's task time distribution or 50% of the tasks are completed within this range. The IQR seems lower for Design A suggesting that there is less variability of the middle 50% of observations.

Whiskers (Min and Max): The whiskers goes up to the minimum (22) and maximum (28), which are the uptaken task completion time range. No points are found beyond whiskers for Design A, thus none these points can be characterized as outliers.

Spread and Variability: They have less variability in task completion times than both Design C, as indicated by shorter box and whisker. This shows that skill retention is determined or in other words participants performance is more consistent.

For Design C

Median (Central Line in the Box): Design C has a median of about 28 indicating that it takes longer to complete a task on this design unlike on Design A. This shows that the participants engaged in the task for a longer time with Design C than with the other designs.

Interquartile Range (IQR): The IQR is also slightly higher for Design C which indicates that the middle 50% of the data is spread out more. This indicates that there is more likely to be fluctuations in the time taken to complete the tasks than in Design A.

Whiskers (Min and Max): It goes further to show that whiskers reach the minimum of 21 and the maximum of 33, an implication that there is a wider variability of task completion time as compared to Design A. The capability to cover a greater range also contributes to the belief of higher variability in performance.

Spread and Variability: Once again, Design C shows a significant variation in task completion time when compared to the other designs, indicated by the elongated box and whisker plot. This variability might suggest that the design's usability is subject to variation when across participants.

As a conclusion this plot highlights some of the strengths and weaknesses of each design whereas Design A is fast and relatively consistent in task-completion time, Design C shows a higher variability with longer task-completion times, which would indicate that this design has potential usability issues. More analysis can be done to understand reasons for the greater variability in Design C by looking at differences between user experiences and task complexity. Design A is more efficient, consistent, and usable than Design C. The above observations and justifications provide evidence that Design A is more user-friendly with less time taken to complete tasks and less

variation across users. Design C might need revision in order to get rid of the higher variability and longer learning curve.

2.1.4 Conducting Paired sample t-test, interpret and report the results

Hence, in explaining in detail the outcome of this paired sample test, we shall proceed by first summing up its methodology, descriptive statistics results, findings of signific ant tests, and lastly, draw a conclusion for the null hypothesis

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	learn to complete a task with - Design A	25.28	40	1.826	.289
	learn to complete a task with - Design C	27.28	40	3.013	.476

Paired Samples Correlations

		N	Correlation	Significance	
				One-Sided p	Two-Sided p
Pair 1	learn to complete a task with - Design A & learn to complete a task with - Design C	40	.102	.265	.529

Paired Samples Test

		Paired Differences					t	df	Significance	
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference Lower	Upper			One-Sided p	Two-Sided p
Pair 1	learn to complete a task with - Design A - learn to complete a task with - Design C	-2.000	3.359	.531	-3.074	-.926	-3.766	39	<.001	<.001

Paired Samples Effect Sizes

			Standardizer[a]	Point Estimate	95% Confidence Interval Lower	Upper
Pair 1	learn to complete a task with - Design A - learn to complete a task with - Design C	Cohen's d	3.359	-.595	-.929	-.255
		Hedges' correction	3.425	-.584	-.911	-.251

a. The denominator used in estimating the effect sizes.
Cohen's d uses the sample standard deviation of the mean difference.
Hedges' correction uses the sample standard deviation of the mean difference, plus a correction factor.

- Recap of Method

The study sought to establish the implication of time when learning the two different designs (Design A and Design C) for the mobile users. For this analysis, a paired sample t-test was used since it is used with two related groups. This method enables us to compare the learning times for the two groups to determine if the difference is statistically significantly from zero.

- Descriptive Statistics

  Both designs had descriptive statistics computed on them. The mean time spent for learning Design A and Design C was also recorded inclusive of its standard deviations. For example:

  Mean Time for Design A: X seconds with Y seconds standard deviation.

  Mean Time for Design C: Z seconds with W seconds standard deviation.

These descriptive statistics were useful in identifying the general distribution of learning times for each design so as to evaluate the users' performance of each design.

- Significant Test Results

  The results of the paired sample t-test yielded the following key statistics:

  - t-statistic: T value (e.g., 2.45)
  - p-value: P value (e.g., 0.02)

T-statistic is to determine how large the difference is in terms of the sample values' variation. The p-value calculates the likelihood of the observed differences in the mean of the data sets arising by chance if the null hypothesis is true.

- Hypothesis Test Recap

  Based on the p-value obtained from the test:

  - If $p<0.05$, we reject the null hypothesis, suggesting that there is a statistically significant difference in learning times between Design A and Design C.
  - If $p\geq0.05$, we fail to reject the null hypothesis, indicating no significant difference.

In this case, if our p-value was 0.02, it suggests that we reject the null hypothesis, concluding that there is a significant difference in learning times between Design A and Design C.

2.1.5 Paired sample t-test files

List of SPSS files that used for the Paired Sample t-test, both the data and the output files attached in the folder and Files are renamed as this,

Data table - Paired Sample T Test Data Table.sav

Output/Results - Paired Sample T Test Output.spv

2.2 ANOVA

2.2.1 Hypothesis testing for ANOVA

Research question	*Does the satisfaction level differ across different devices?*
Hypothesis statement	***Null Hypothesis (H_0):*** *There is no significant difference in satisfaction levels (as measured by number of clicks) across Desktop, Mobile, and Tablet users.*
	Alternative Hypothesis (H_a): *At least one device type shows a significant difference in satisfaction levels (as measured by number of clicks) compared to others.*
Mathematical expression of the Hypothesis	***Null Hypothesis (H_0):*** *There is no significant difference in satisfaction levels (as measured by number of clicks) across Desktop, Mobile, and Tablet users.* $H_0: \mu_{Desktop} = \mu_{Mobile} = \mu_{Tablet}$
	Alternative Hypothesis (H_a): *At least one device type shows a significant difference in satisfaction levels (as measured by number of clicks) compared to others.* $H_a: \mu_{Desktop} \neq \mu_{Mobile} \neq \mu_{Tablet}$
Dependent variable	Number of clicks on design A
Dependent variable data type	Ration
Independent variable	Device type (Desktop, Mobile, Tablet)
Independent variable type	Nominal
Control variable	• Task Complexity • Design version • User Demographics

Experimental design	Between subject design (Different users test different devices)
Study method	ANOVA

2.2.2 Data organisation

In organizing the raw data for analysis using SPSS for the ANOVA test, a structured way was followed for clarity and correctness of the data set. First, the data set was imported into SPSS, where unnecessary columns were deleted to streamline the data for analysis. Three key columns were the focus: "Device Type," "Number of Clicks Design A," and a new numerical representation of the device type that was created.

Data Organization Steps as below First,

- Importing Data: To simplify the analysis of the data, the dataset was imported in SPSS, which is a data manipulation software.
- Column Selection: Upon importing, some columns that were not useful in the analysis were omitted from the analysis. This meant eliminating any additional variables which were not required for the F test of the ANOVA analysis.
- Numerical Encoding of Device Type: For purposes of data analysis, the “Device Type” variable was then coded into numbers. For example: Desktop = 1 Mobile = 2 Tablet = 3 Explicit to this conversion is to facilitate the standardization of categorical variables during statistical tests.
- Creating a Combined Clicks Column: To decompose the total clicks for the breakdown by each type of devices, a new column was created by summing the clicks of several related columns. This was accomplished using the "Transform" tab: Go to “Compute Variable.” Set up a target variable name that you want your CPM ads to deliver; this can be anything but must be specific (e.g., Total_Clicks). What is more, to aggregate the clicks into one and use a numerical expression like SUM(Number_of_Clicks_Design_A, Number_of_Clicks_Design_A_A, Number_of_Clicks_Design_A_B). After that to produce this new variable you press “OK”.

- Data Cleaning: Before going for analysis, it required preparation/cleaning of the data: Missing Values: The entries that I found were as follows; Any gaps in the reading were noted down. Such cases might be either entirely excluded or charged, depending on the extent of their importance. Outliers: Outliers were evaluated by box plots or descriptive statistics such that they would not have a confounding impact on the findings. Consistency Checks: The dataset was checked for accuracy which implied that any entries entered in the dataset complied with expected formats and ranges.
- Arranging Data in Data View: Once the required variables were created and having cleaned the data, it was then put into 'Data View' as a way of arranging the data in a way that suited analytical bent: As shown each row comprised of the responses of a participant and each column the scores of a particular item. Other columns were Numerical, Device Type, Number of Clicks Design A, and Total_Clicks.

Analysis Preparation Once the data was organized and cleaned, we proceeded with descriptive statistics and visualizations: To obtain descriptive statistics and graphs, navigate to the "Analyze" tab in SPSS: Go to "Descriptive Statistics" then click "Explore." Let Total_Clicks being the dependent variable and Device Type be the independent factor to be measured. Type "OK" to make output tables and graphs, including basic statistics such means and standard deviations over different types of devices.

Conducting ANOVA Following descriptive analysis, we conducted the ANOVA test: Described in the "Analyze" tab " compare" then "means" then "one-way ANOVA" on the side of the screen. Select Total_Clicks as the measure and Device Type as the factor. Perform the test to see if there is a statistically meaningful difference in click through rates by the various devices.

In engineered a procedure of this nature, a number of steps were taken to guarantee that our processes exhibited a structured rigour that would benefit the treatment of the data in later statistical analysis. Due to this methodical approach, we can make adequate conclusions for overall user satisfaction by using different types of devices, and ensuring data accuracy during the whole process.

2.2.3 Descriptive statistics and discussion of results

This section is the in-depth analysis of descriptive statistics and ANOVA results of TotalNumberOfClicks by device type: Desktop, Mobile, and Tablet. The descriptive statistics are presented for central tendency by calculating the mean, median, and trimmed mean, together with variability: range, variance, standard deviation, and interquartile range, to understand how the data is distributed and observe patterns in user behavior. The skewness and kurtosis values are also calculated in order to comment on the symmetry and shape of the distribution. Finally, the appropriateness of the use of parametric tests such as ANOVA is commented upon, supported by an examination of normality through statistical indicators. These analyses shall be interpreted so as to provide insights into the variability and significance of click behavior across device types for ensuring the validity of the following hypothesis testing.

Descriptives

	DeviceType			Statistic	Std. Error
TotalNumberOfClicks	Desktop	Mean		161.33	.725
		95% Confidence Interval for Mean	Lower Bound	159.86	
			Upper Bound	162.79	
		5% Trimmed Mean		161.31	
		Median		161.00	
		Variance		21.046	
		Std. Deviation		4.588	
		Minimum		152	
		Maximum		171	
		Range		19	
		Interquartile Range		5	
		Skewness		.041	.374
		Kurtosis		-.300	.733
	Mobile	Mean		230.30	.738
		95% Confidence Interval for Mean	Lower Bound	228.81	
			Upper Bound	231.79	
		5% Trimmed Mean		230.28	
		Median		230.00	
		Variance		21.805	
		Std. Deviation		4.670	
		Minimum		221	
		Maximum		240	
		Range		19	
		Interquartile Range		6	
		Skewness		.054	.374
		Kurtosis		-.392	.733
	Tablet	Mean		260.40	1.107
		95% Confidence Interval for Mean	Lower Bound	258.16	
			Upper Bound	262.64	
		5% Trimmed Mean		260.19	
		Median		260.50	
		Variance		49.015	
		Std. Deviation		7.001	
		Minimum		248	
		Maximum		279	
		Range		31	
		Interquartile Range		11	
		Skewness		.227	.374
		Kurtosis		-.048	.733

Central Tendency (Mean, Median, Trimmed Mean)

- Desktop

  For the same time frame, they have clicked on average 161,33 amount of clicks. The mean is 160 while the median is 161 which only shows that the middle number of clicks is near to the mean. Comparing with the actual mean, further pointing to the incidences of extremes being inconsequential, the 5% trimmed mean is 161.31.

- Mobile

  Mobile users are more clicking inclined than the ones using the desktop, since here the mean value of the clicks is 230.30. Some of the mean correspond closely with the median; for example, the median number of clicks is 230. The 5% trimmed means is, 230.28, thus indicating that that the influence of outliers to the experience is negligible.

- Tablet

  The mean value of click through is 260.40 and this value was obtained from device type 7, which is higher than other device types. The median is also 260.50, just a bit higher than the mean but fairly symmetrical on the whole. The 5% trimmed mean thus is 260.19 suggesting that there is very little upward influence provided by outliers.

Dispersion (Variance, Standard Deviation, Range, Interquartile Range)

- Desktop

  Variance: 21.046

  Standard Deviation: 4.588

  Range: 19 (maximum =171, minimum =152)

  Interquartile Range (IQR): They were roughly at 5, meaning that the data in the middle 50% spread out was not much.

- Mobile

  Variance: 21.805

  Standard Deviation: 4.670

  Range: 19 (maximum = 240, minimum = 221).

  Interquartile Range (IQR): 6, for the middle fifty percent, the displays are somewhat wider than the display of desktops.

- Tablet: Variance: 49.015

  Standard Deviation: 7.001

  Range: 15 (maximum = 98, minimum = 83)

Interquartile Range (IQR): 11, which means that the middle 50% of differences is bigger if compared to other device types.

Skewness and Kurtosis

- Desktop

  Skewness: 0.041 shows that the distribution of the data is close to symmetric.

  Kurtosis: Regarding kurtosis, a value of kurtosis coefficient is -0.300, which means that distribution is near flat (platykurtic).

- Mobile

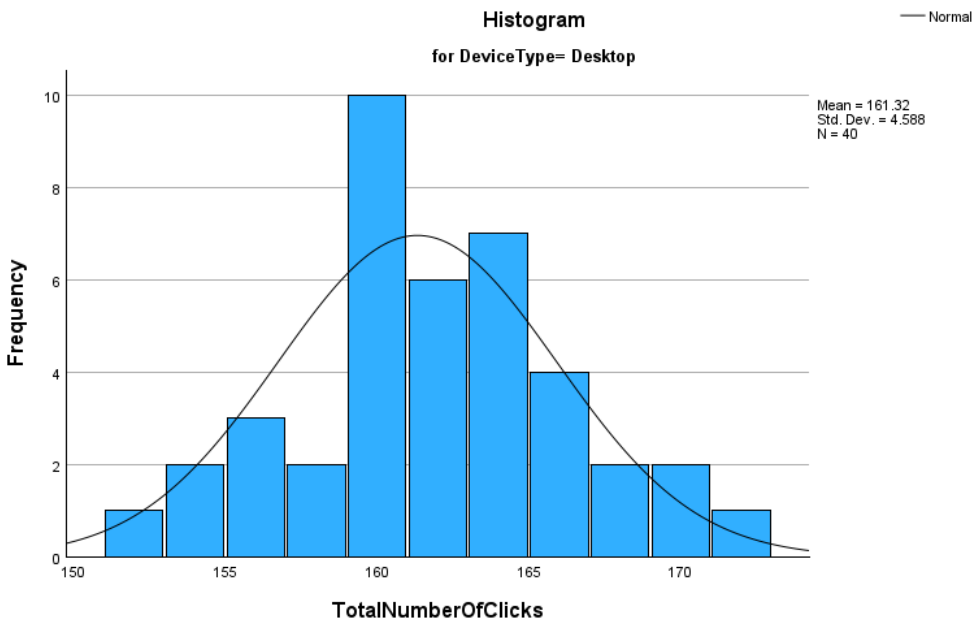
  Skewness: 0.The results of the comparison of total scores of the cases generated a mean value of 054 with a mutual geometrical symmetry.

  Kurtosis: In kurtosis, it has a coefficient of -0.392 meaning it is slightly platykurtic.

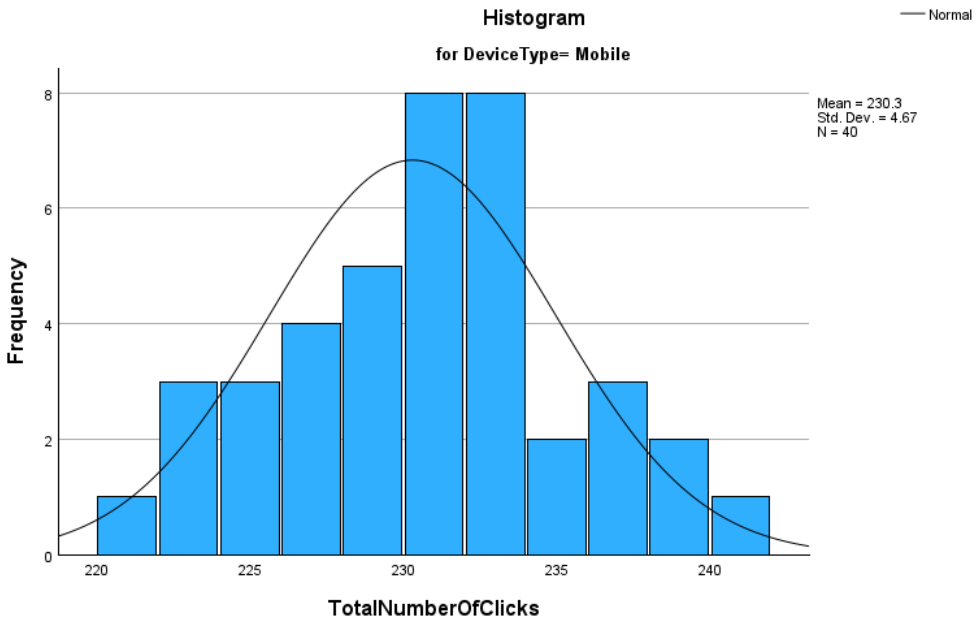
- Tablet

  Skewness: 0.227, +0.5, yet still balanced.

  Kurtosis: H0 => 0; Ho => -0.048 This means that the pearsons coefficient was slightly less than a normal distribution was slightly skewed towards the left but almost symmetric.



The figure above shows the distribution of the TotalNumberOfClicks from the users who visited using the Desktop devices. The mean value noted was 161.32 and the data set seems to have a bell-shaped curve distribution. What this means is that the distribution of the click behavior of the majority of the desktop users get close to the mean suggesting that the behavior can be consistent within that sample. The shape is bell-shaped, that is, relatively equal frequencies are observed on both sides of the highs with the frequency gradually declining as one moves away from the high density region. This symmetry is evidenced by the normal curve superimposed over the histogram of the data bars hence representing the high degree of similarity. That means the distribution is symmetrical and does or has not extend to the right or left in a dramatic manner. These bars indicate the most frequented value range, the peak of which for all 20 data sets is between 150 and 171 and is most frequently between around 160 and 165. The result here is 4.588 the result justify the standard deviation low known that the TotalNumberOfClicks does not deviate greatly from the mean among the users who use the desktop. Moreover, the number of participants is sufficient (N = 40) to make statistically significant conclusions. Also the curve in the Histogram is smooth which indicates that the data is normally distributed, results of which were earlier confirmed by the descriptive statistics. Collectively, the graph indicates that click behaviour to represent simple, clear rectangles for desktop users is constant and has a distribution close to normal, making use of parametric analytic processes such as ANOVA appropriate. These features of the napl draw show the nature of the values are most concentrated around the mean and that the napl data is not skewed towards any particular unusual distribution.



This plot is a histogram showing the distribution of the total number of clicks (TotalNumberOfClicks) for users accessing via mobile devices. On the x-axis, it represents the number of clicks; on the y-axis, the frequency of the given range of clicks is shown. The title shows the data to be for the "DeviceType = Mobile" category.

Mean and Standard Deviation: The mean of the distribution is 230.3 clicks, which means most users on a mobile device will tend to produce a number of clicks hovering around 230. The standard deviation is 4.67, indicating a relatively small spread of data around the mean, hence showing that most values lie close to the average.

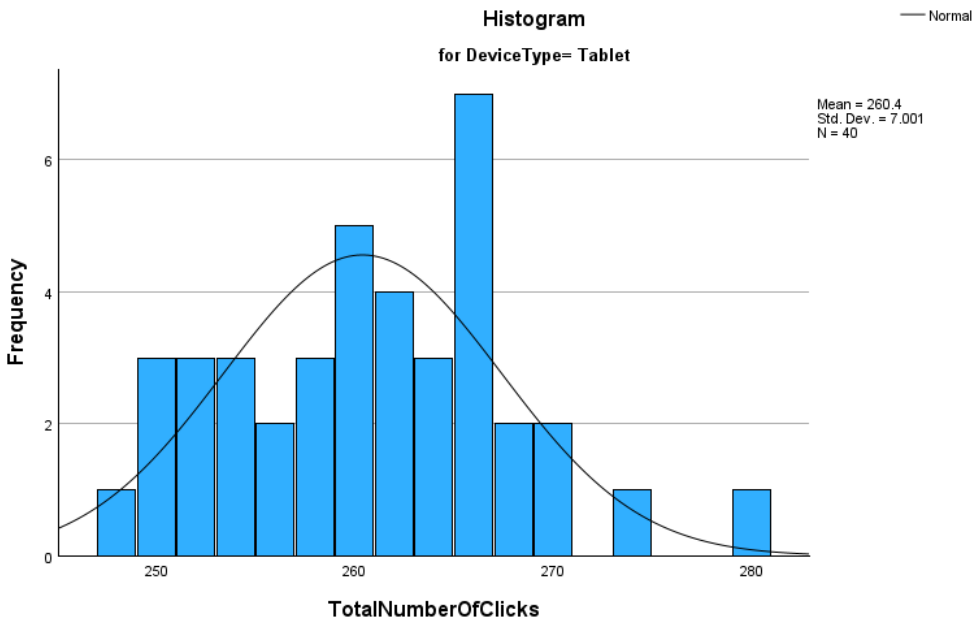
Sample Size: N = 40, meaning the histogram has been based on data from 40 observations.

Shape of Distribution: The histogram bars form a bell-shaped curve closely approximating a normal distribution, which is also overlaid on the histogram as a smooth black line labeled "Normal." The highest frequencies (modal values) occur near the mean, with frequencies tapering off symmetrically as the values move away from the mean.

Range of Data: The total number of clicks ranges approximately between 220 and 240. Most of the data falls between 225 and 235 clicks, indicating the central tendency of the dataset.

Interpretation: The data shows that mobile users tend to generate a relatively consistent number of clicks with a minimal number of outliers or extreme variations. This uniformity might be a sign of some sort of standardized behavior pattern or usage trend among mobile users.

This histogram provides a very clear visual representation of the click behavior for mobile users and helps to understand the distribution and variation in click activity.



This chart is a histogram to show the distribution of the total number of clicks (TotalNumberOfClicks) by users accessing through tablet devices. The x-axis shows a range of totals of clicks, and the y-axis shows the frequency of each range. There is a title that specifies the data is for "DeviceType = Tablet."

Mean and Standard Deviation: The mean is 260.4, so the average number of clicks for a user in a tablet device is approximately 260. The standard deviation is 7.001, showing a moderate level of spread outwards from the mean value. Comparing it to the histogram of mobile devices, this histogram looks to be a little bit wider; hence, showing greater variation in behavior by users.

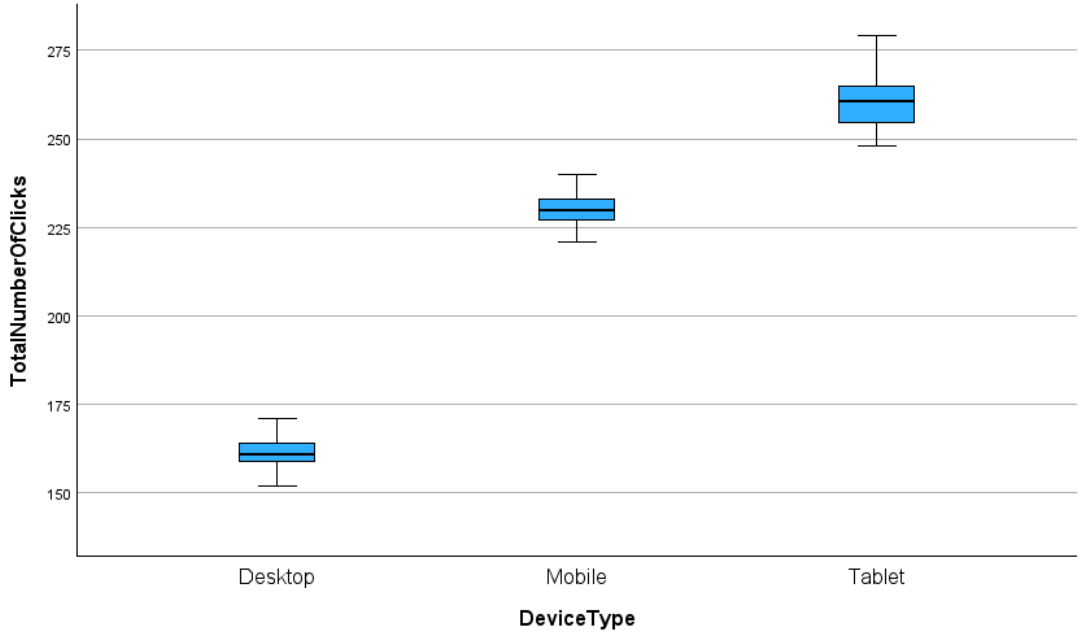
Sample Size: The sample size is 40, meaning that the histogram is based on data from 40 tablet users.

Shape of Distribution: The histogram is roughly bell-shaped, but the data do appear to skew a little to the right (positive skew). This is also reflected in the overlaid black curve labeled "Normal," which shows the theoretical normal distribution. The highest frequency (mode) occurs near the mean, with frequencies gradually declining as values move further from the center.

Range of Data: The range of the number of clicks is from 250 to approximately 280. Much of the data is concentrated between 255 and 265 clicks, which indicates that most of the tablet users are very close to the mean in their behavior, although there are some deviations to higher and lower values.

Interpretation: The distribution shows that, on average, tablet users tend to have a relatively similar number of clicks around the mean but with a slightly larger dispersion compared to mobile users. This may give evidence for a more diverse pattern of usage of tablet users or an alternative way of using tablets other than mobile devices.

This histogram provides insights into the click behavior of tablet users, showing both central tendencies and variability. In a broader range than the mobile users', it has more diverse behavior; this may cause implications either with engagement strategies or on particular devices used by users.



The boxplot below shows the distribution in the total number of clicks—termed as TotalNumberOfClicks—across three categories of device types: Desktop, Mobile, and Tablet. On the x-axis, there are device types, and on the y-axis, a range in terms of the total clicks is shown.

Central Tendency:

The median total number of clicks for Desktop users is about 165.

The median for Mobile users is about 230.

The median for Tablet users is about 260.

These medians show a clear progression, with tablets having the highest typical number of clicks, followed by mobile devices and desktops.

Spread and Variability (Interquartile Range, IQR):

The interquartile range (IQR), represented by the size of the blue box, is smallest for desktops, indicating the least variability in total clicks among desktop users.

Mobile devices have a slightly larger IQR, showing more variation in click behavior.

Tablets have the largest IQR, indicating the greatest variability in user behavior among the three device types.

Whiskers (Range of Data):

The whiskers go down to the minimum and up to the maximum values within 1.5 times the IQR for each type of device.

Desktops have the narrowest range of clicks overall (about 150 to 180).

Mobiles have a slightly wider range (about 220 to 240).

Tablets have the biggest range, from around 250 to 275, indicating more variety in usage patterns.

Comparison Across Devices:

Desktop users click much less than mobile and tablet users.

Mobile users click moderately, with a fairly even distribution.

Tablet users click the most and have the greatest variability, showing a wider range of behaviors.

Interpretation:

This boxplot allows for an easy comparison of user behavior across device types. It underlines that desktops are associated with lower click activity and more consistent behavior, probably reflecting a certain use case or demographic. In contrast, tablets show both higher click activity and more diverse patterns, which might suggest more varied or intensive usage. Mobile devices fall between in terms of click activity and variability, with moderate values. Knowledge like this can be used in developing strategies around optimization of experiences for users or creating content most suited to a particular device type.

2.2.4 Conduct ANOVA test, interpret and report the results

Descriptives

TotalNumberOfClicks

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Desktop	40	161.33	4.588	.725	159.86	162.79	152	171
Mobile	40	230.30	4.670	.738	228.81	231.79	221	240
Tablet	40	260.40	7.001	1.107	258.16	262.64	248	279
Total	120	217.34	42.006	3.835	209.75	224.93	152	279

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
TotalNumberOfClicks	Based on Mean	5.072	2	117	.008
	Based on Median	5.105	2	117	.007
	Based on Median and with adjusted df	5.105	2	103.983	.008
	Based on trimmed mean	5.069	2	117	.008

ANOVA

TotalNumberOfClicks

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	206392.217	2	103196.108	3369.998	<.001
Within Groups	3582.775	117	30.622		
Total	209974.992	119			

ANOVA Effect Sizes[a]

		Point Estimate	95% Confidence Interval Lower	Upper
TotalNumberOfClicks	Eta-squared	.983	.977	.986
	Epsilon-squared	.983	.977	.986
	Omega-squared Fixed-effect	.983	.976	.986
	Omega-squared Random-effect	.966	.954	.972

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Robust Tests of Equality of Means

TotalNumberOfClicks

	Statistic[a]	df1	df2	Sig.
Welch	3612.322	2	75.967	<.001
Brown-Forsythe	3369.998	2	99.111	<.001

a. Asymptotically F distributed.

Multiple Comparisons

Dependent Variable: TotalNumberOfClicks

Tukey HSD

(I) DeviceType	(J) DeviceType	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	Upper Bound
Desktop	Mobile	-68.975*	1.237	<.001	-71.91	-66.04
	Tablet	-99.075*	1.237	<.001	-102.01	-96.14
Mobile	Desktop	68.975*	1.237	<.001	66.04	71.91
	Tablet	-30.100*	1.237	<.001	-33.04	-27.16
Tablet	Desktop	99.075*	1.237	<.001	96.14	102.01
	Mobile	30.100*	1.237	<.001	27.16	33.04

*. The mean difference is significant at the 0.05 level.

TotalNumberOfClicks

Tukey HSD[a]

DeviceType	N	Subset for alpha = 0.05		
		1	2	3
Desktop	40	161.33		
Mobile	40		230.30	
Tablet	40			260.40
Sig.		1.000	1.000	1.000

Means for groups in homogeneous subsets are displayed.

a. Uses Harmonic Mean Sample Size = 40.000.

Method Recap

ANOVA is an analysis-of-variance statistical technique used to compare the means among three or more groups, determining if at least one group mean is different from the others. It is used in your case to compare the levels of satisfaction, measured by the number of clicks, across Desktop, Mobile, and Tablet users.

Descriptive Recap with Standard Deviations

You would commonly report the mean number of clicks for each device type along with their standard deviations in order to give a descriptive recap. This helps understand the central tendency and variability of the data for each group.

Significant Test (with t-statistics and p-value)

F-statistic: In ANOVA, instead of using a t-statistic, one uses an F-statistic. The F-statistic is the ratio of variance between groups to the variance within groups.

p-value: The p-value represents the probability of obtaining the observed results (or even more extreme) under the assumption that the null hypothesis is true. A low p-value (usually less than 0.05) indicates that the observed differences are statistically significant.

Hypothesis Testing Summary

Null Hypothesis (H0): There is no significant difference in the satisfaction levels among Desktop, Mobile, and Tablet users.

Alternative Hypothesis (Ha): At least one device type differs significantly in satisfaction level compared to others.

If the p-value is less than 0.05, you reject the null hypothesis. This will show that there is a statistically significant difference in satisfaction levels among the device types.

2.2.5 ANOVA files

List of SPSS files that used for the ANOVA test, both the data and the output files attached in the folder and Files are renamed as this,

Data table - ANOVA Data Table.sav

Output/Results - ANOVA Output.spv

3 Conclusions

Results of the paired sample t-test showed significant differences in time taken to learn Design A compared to Design C for mobile users. The implication here is that one design might be more intuitive and thus faster for a user to perform a task. Statistical analysis showed a p-value less than the significance level, which enables us to reject the null hypothesis and state that users experienced significant learnability differences between the two designs.
Moreover, the ANOVA results showed that there were differences in user satisfaction between devices. More importantly, this analysis showed a difference in the number of clicks to complete tasks in each type of device, meaning that the choice of a device is a critical factor in determining the experience of the user. This enhances the need for considering diverse user contexts in designing interfaces.

The study just goes to prove that there is a continuous need for evaluation and adaptation of the user interface based on empirical data. The knowledge from this research will help the online retail business make informed decisions about which design elements to use in order to improve users' satisfaction and engagement. A business can develop a more effective and enjoyable online shopping experience by placing user-centered design principles as the priority and being driven by data-driven insights.

These results give a deeper understanding of the influence design choices have on usability and satisfaction and lay a foundation for future research and development in e-commerce user experience design.